# Zensar Content Studio

## Project Documentation

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## 1. Purpose

Teams spend a lot of time turning documents into presentations, videos, publisher-ready papers and branded white papers. Zensar Content Studio does this automatically, with three rules:

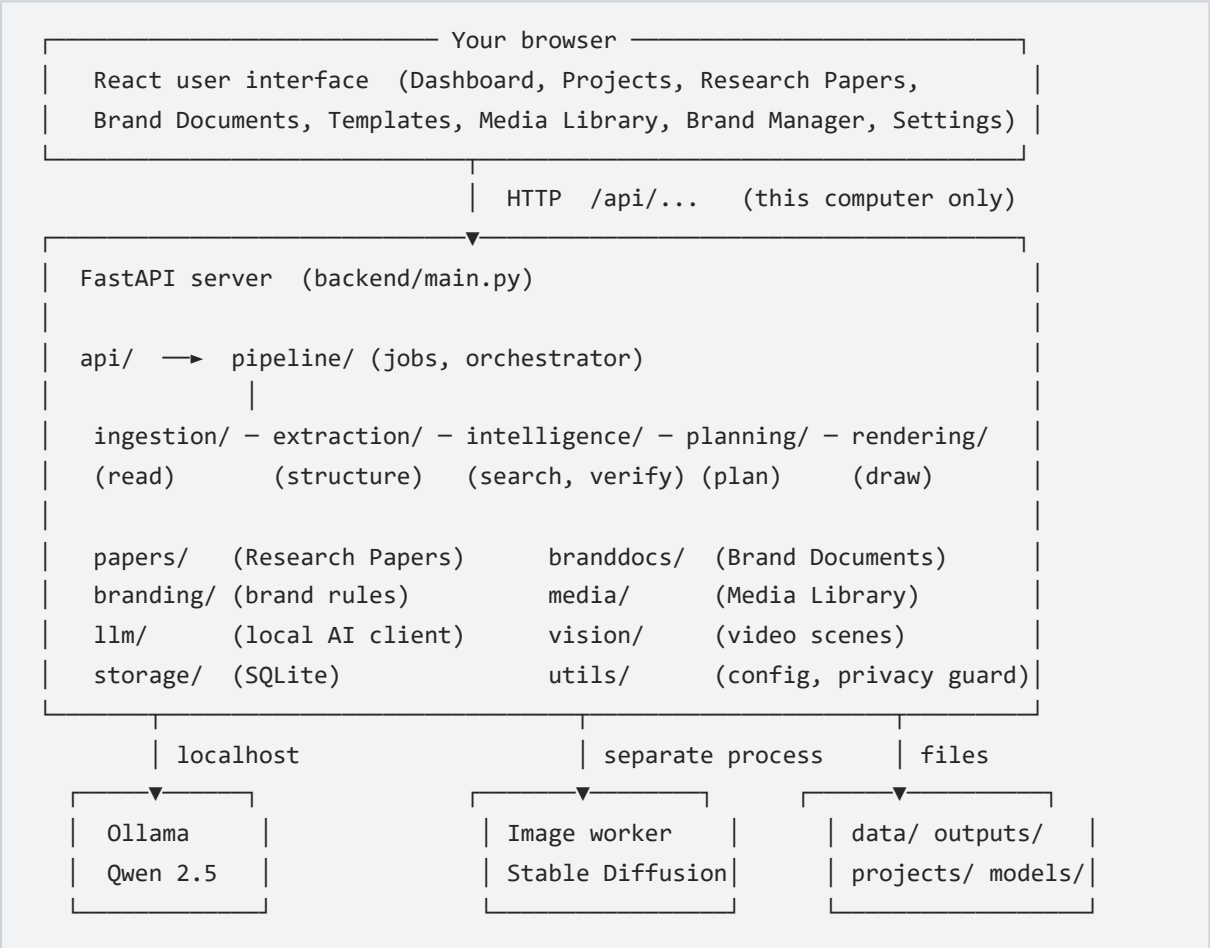
1. **Private:** everything runs on the user's own computer. No document is ever sent to a cloud service.
2. **Faithful:** every statement in the output is traced back to the source. Numbers are checked, and nothing is invented.
3. **On-brand:** the design comes from the brand's own official material, never from guesses.

## 2. The three tools

| Tool | Input | Output |
|---|---|---|
| **Projects** | Documents (PDF, Word, PowerPoint, text, web pages), demo videos, code repositories | Editable PowerPoint, PDF, brand-styled images and diagrams, a narrated MP4 video |

| Tool | Input | Output |
|---|---|---|
| **Research Papers** | A scientific paper (PDF, Word, LaTeX, Markdown, text) | The paper re-formatted for **Springer Nature, Springer LNCS, Elsevier, IEEE, ACM or APA 7**: LaTeX project, Word, PDF, BibTeX and a checks report |
| **Brand Documents** | Any document (PDF, Word, Markdown, text) plus a branded **reference PDF** | A branded white paper (PDF) laid out like the reference, fitted to its length, with images created for the content |

## 3. Architecture at a glance




- The **interface** only displays things and sends requests. All the work happens in the **server**.
- Long tasks run as **background jobs**. The interface checks their progress every couple of seconds.
- The **writing AI** (Ollama) and the **image AI** (Stable Diffusion) run as local programs. The image AI runs in its own process, so its memory is freed after each job.
- A **network guard** blocks every outgoing connection except to this computer.

# 4. Technology

| Area | Technology | Used for |
|---|---|---|
| Interface | React 18, TypeScript, Tailwind CSS 4, Vite 6, lucide icons | Screens in the browser |
| Server | Python 3.10–3.12, FastAPI, Uvicorn, Pydantic | API, background jobs, data checking |
| Storage | SQLite + files | Projects, jobs, settings, outputs |
| Reading documents | PyMuPDF, python-docx, python-pptx, BeautifulSoup, lxml, RapidOCR (optional) | PDF, Word, PowerPoint, HTML, scanned pages |
| Meaning search | all-MiniLM-L6-v2 (ONNX Runtime + tokenizers), FAISS | Finding relevant passages |
| Writing AI | **Qwen 2.5** (7B or 3B) via **Ollama** | Planning, condensing, scene descriptions |
| Image AI | **DreamShaper 8 LCM** (Stable Diffusion 1.5) via Diffusers + PyTorch | Pictures for brand documents |
| Vision AI | qwen2.5vl:3b via Ollama (optional) | Describing demo-video scenes |
| Speech | Windows SAPI voices (optional Piper) | Video narration |
| Output | python-pptx, ReportLab, Pillow, Matplotlib, FFmpeg (imageio-ffmpeg), fontTools | PowerPoint, PDF, images, charts, video, fonts |
| Papers | citeproc-py + official CSL styles, bibtexparser, pylatexenc, Jinja2 | Citations, references, LaTeX |
| Testing | pytest (127 tests) | Automatic quality checks |
| Setup | `start.bat` → PowerShell `bootstrap.ps1` | One-click install and start |

## AI models

| Model | Size | Licence | Runs on |
|---|---|---|---|
| Qwen 2.5 7B / 3B | 4.7 / 1.9 GB | Apache-2.0 | Ollama (GPU if available) |

| Model | Size | Licence | Runs on |
|---|---|---|---|
| DreamShaper 8 LCM | 2.1 GB | CreativeML OpenRAIL-M (commercial use allowed) | GPU (CUDA) or CPU |
| all-MiniLM-L6-v2 (ONNX) | 90 MB | Apache-2.0 | CPU |
| qwen2.5vl:3b (optional) | 3.2 GB | Apache-2.0 | Ollama |

## 5. Privacy and trust

| Protection | How it works |
|---|---|
| No uploads | `utils/network_guard.py` refuses any outgoing connection that isn't to this computer. |
| Local AI only | `llm/ollama.py` refuses AI servers that are not on this computer. |
| No document text in logs | `utils/logging.py` records only counts and timings. |
| No invented facts | `intelligence/verification.py` checks every statement against its source; numbers must match exactly. Brand-document condensing ( `branddocs/fit.py` ) rejects AI text containing numbers that are not in the source. |
| No invented brand values | `branding/brand_profile.py` and `branding/theme.py` keep unknown values empty and show neutral fall-backs. |
| Optional online features | GitHub import and licensed image search work only when the user asks. They use an allow-list of hosts and never send document text. |

## 6. Folders

| Folder | Contents |
|---|---|
| `backend/` | The engine (Python) |
| `frontend/` | The interface (React); `dist/` is the ready-built version |
| `brands/` | Brand profiles and assets (Zensar) |
| `scripts/` | Setup, model download, offline bundle, demo recording, developer checks |
| `samples/` | Example documents, video and papers |
| `tests/` | Automatic tests |

| Folder | Contents |
|---|---|
| `docs/` | This documentation, the slide decks and the demo video |
| *created on first run:* `.venv/` `data/` `outputs/` `projects/` `models/` `logs/` `offline/` | Private Python, user data, results, AI models, logs, offline bundle |

`docs/User_Manual.pdf` explains every file in plain language.

## 7. Installation and deployment

- **Normal PC:** double-click `start.bat`. It installs Python, packages, models, the interface and the local AI, then starts the app at http://localhost:8000.
- **GPU server:** `start.bat` detects the NVIDIA GPU and installs the CUDA build of PyTorch. It picks Qwen 2.5 7B when the GPU has 8 GB or more.
- **Corporate proxy:** a proxy configured in Windows or Git is detected and passed on. If pypi.org is still unreachable, everything is installed from the Zen-TMT-offline bundle over Git.
- **Remote access:** `ZCS_HOST=0.0.0.0` (the app has no login, so restrict access with the firewall), or an SSH tunnel.

## 8. Quality

- **127 automatic tests**: run `python -m pytest`.
- Every conversion ends with a **report**: words preserved, publisher checks, fonts used, images created.
- The documentation in `docs/` is part of the repository and is updated together with the code.

## 9. Related documents

- `docs/User_Manual.pdf` : installation step by step, how to use every screen, every file explained.
- `docs/Pipeline_Explanation.pdf` : how each pipeline works, step by step.
- `docs/Module_Architecture.pptx` : slides on the modules.
- `docs/Pipeline.pptx` : slides on the pipelines.
- `docs/demo/Zensar_Content_Studio_Demo.mp4` : the demo video.